Cranberries and Heat Stress

Yield and Temperature Analysis

Group 5

STAT 628 Data Science Practicum
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WISCONSIN
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Outline

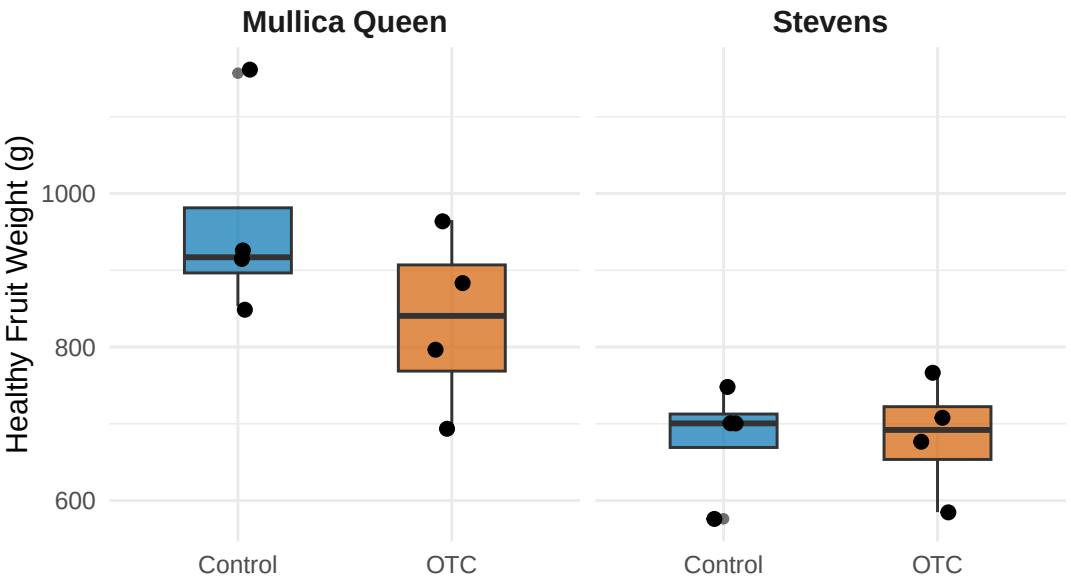
I. **Yield Analysis**
Long-term and acute heat effects on fruit weight and rot

II. **Temperature & Relative Humidity Analysis**
Heat control effectiveness and connection to yield

III. **Conclusions, Limitations & Future Directions**

Yield Analysis: Long-Term Experiment

No significant heat effect on yield

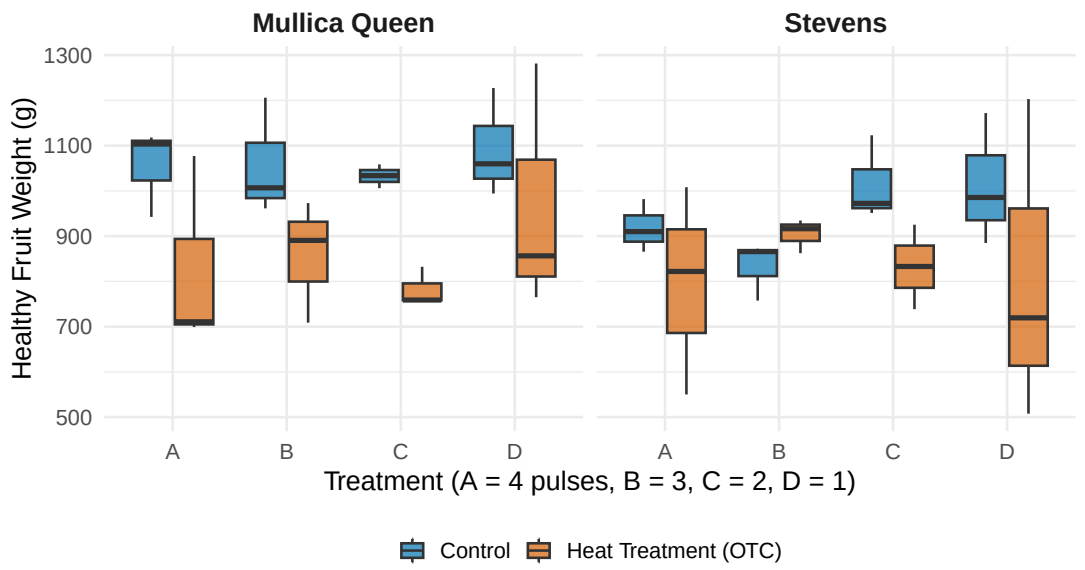


- Mullica Queen produces **280g more** healthy fruit than Stevens ($p = 0.002$).
- OTC heat treatment shows **no significant effect** on yield ($p = 0.19$).
- Only 4 biological replicates per group — limited statistical power.

Model: Blocking ANOVA with set pair as fixed block + permutation test for robustness.

Yield Analysis: Acute Heat Reduces Healthy Fruit Weight

OTC reduces yield by 155g ($p = 0.001$)

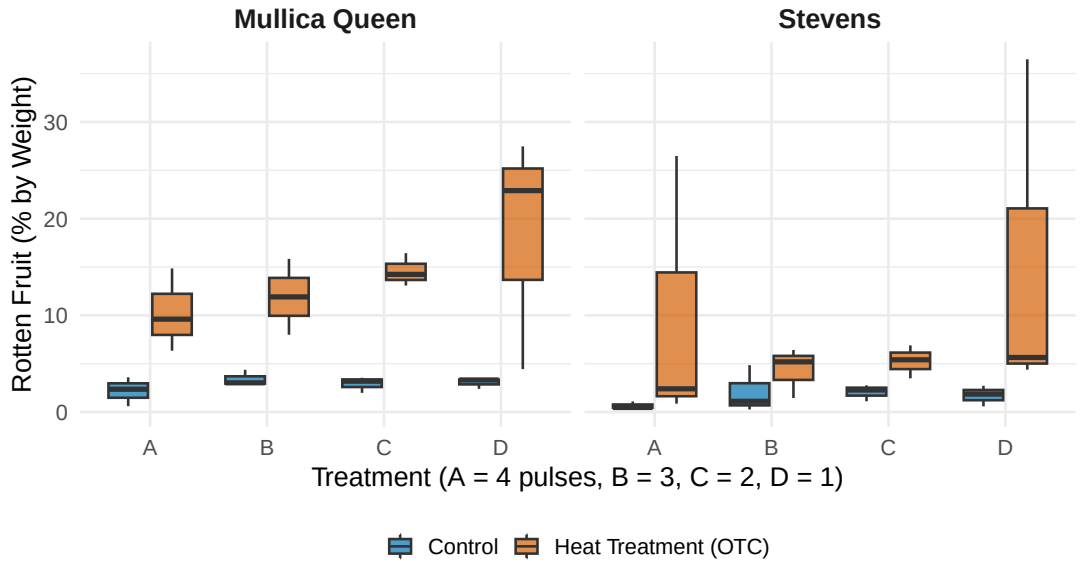


- OTC (orange) consistently yields **less** than Control (blue).
- Mean reduction: **155g** per plot ($p = 0.001$).
- No dose–response: 1 pulse causes the **same damage** as 4 pulses.

Model: Healthy weight $= 1011 - 155 \times \text{Heat} - 69 \times \text{Stevens} + \varepsilon, \quad R^2 = 0.29$

Yield Analysis: Acute Heat Increases Fruit Rot

OTC increases rotten odds by 3.8× ($p < 0.001$)

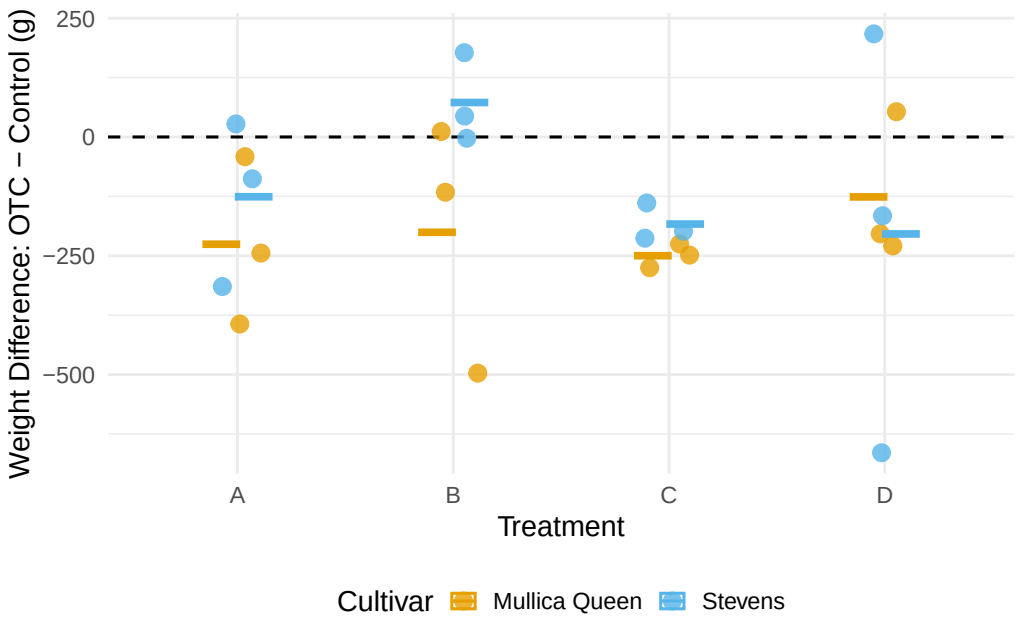


- Control rotten rate: **1–5%**.
- OTC rotten rate: **5–37%**.
- OTC increases odds of rot by **3.8 times**.

Model: Quasibinomial GLM on raw counts — appropriate for proportion data with varying denominators.

Yield Analysis: No Dose–Response

All treatments cause similar yield loss

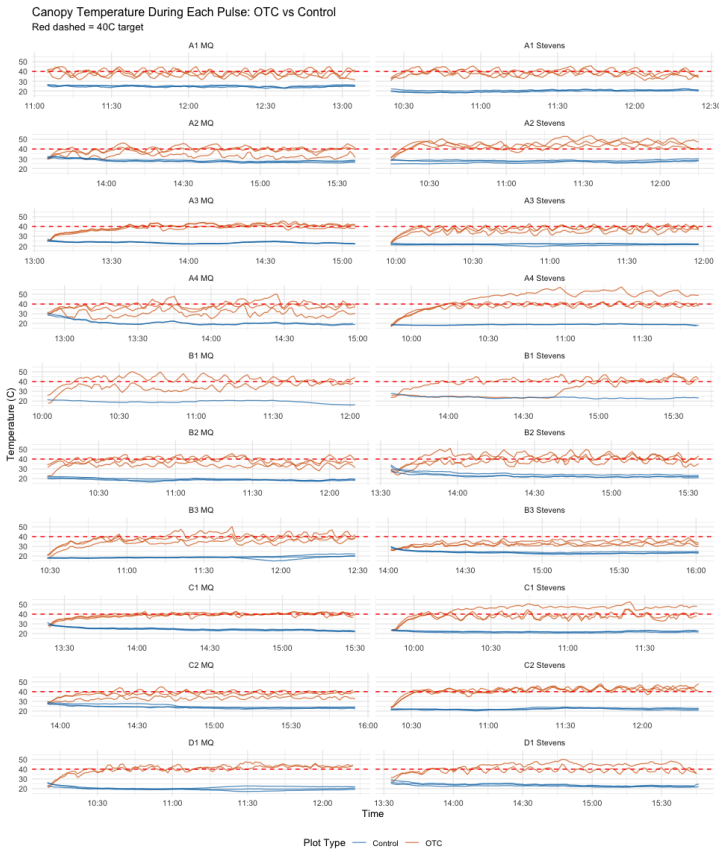


- Each dot = one OTC–Control pair.
- Nearly all dots **below zero**: heat reduced yield.
- Treatment A (4 pulses) ≈ D (1 pulse).
- Suggests a **threshold effect**: even one 2-hour pulse at 35–40°C is enough to cause damage.

Yield Analysis: Summary

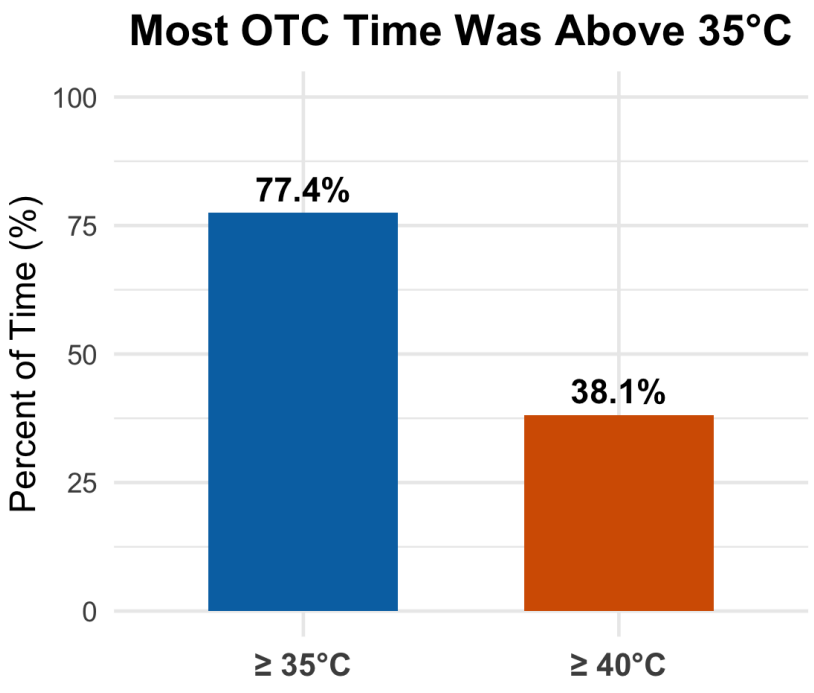
Finding	**Detail**	p-**value**
Long-term OTC	No detectable yield effect	0.19
Acute OTC reduces yield	-155g healthy fruit per plot	0.001
Acute OTC increases rot	Odds ratio $= 3.8\times$	< 0.001
No dose–response	1 pulse $\approx$ 4 pulses	All > 0.2
Cultivar matters	Mullica Queen yields more	0.002

OTC created real heat stress



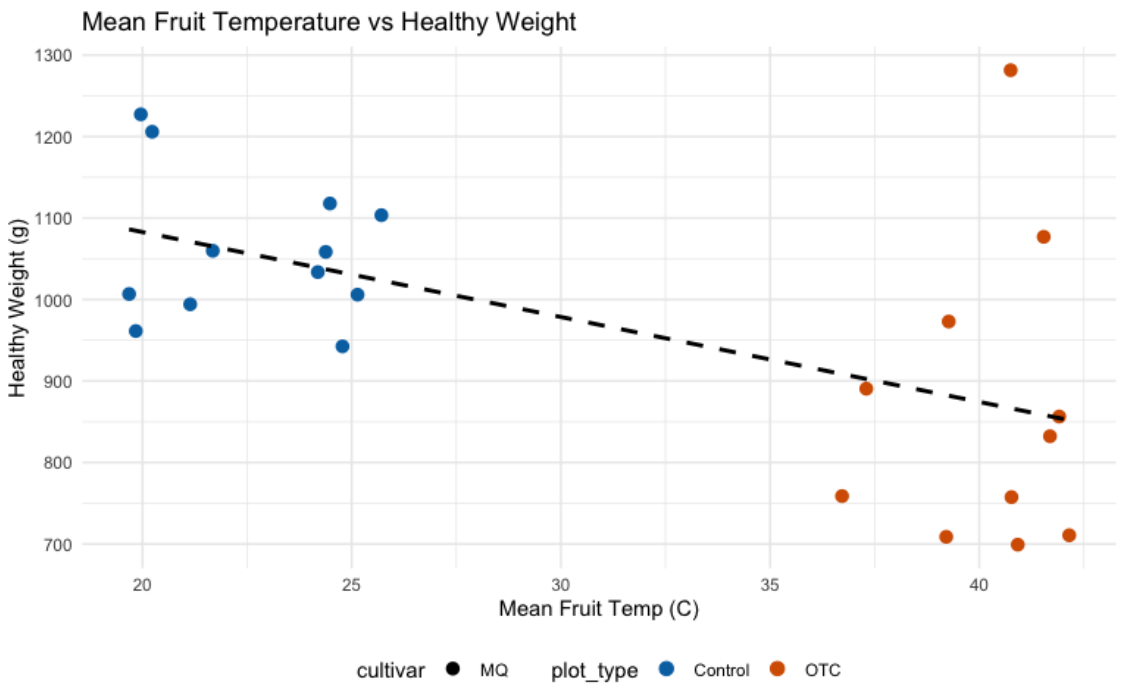
- Control plots stayed around 20–25°C.
- OTC plots often reached 35–45°C.
- The temperature gap was clear across all acute heat pulses.
- This shows that the OTC treatment successfully created strong heat exposure.

Evidence for a temperature threshold



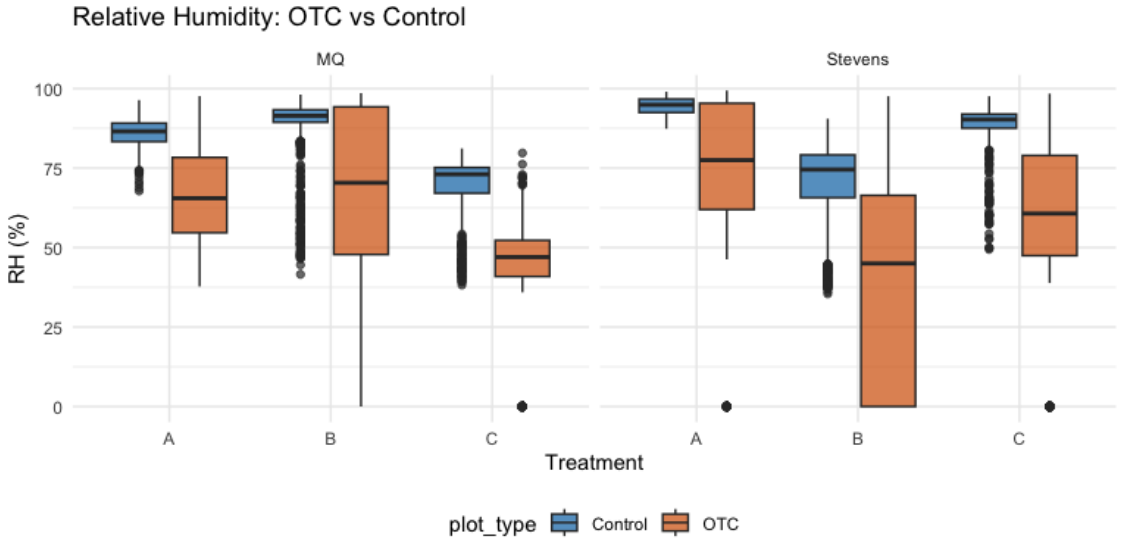
- 40°C was the target, but it was not always maintained.
- Even so, about 77% of OTC time was still above 35°C.
- This suggests that fruit damage may begin around 35°C.
- It also helps explain why treatments A–D showed similar yield results.

Higher fruit temperature was linked to lower yield



- Higher fruit temperature was associated with lower healthy fruit weight.
- The pattern appears in both cultivars.
- This links actual heat exposure to yield loss.
- It supports the acute yield results shown earlier.

OTC also created a hotter and drier environment



- OTC plots had lower relative humidity than control plots.
- This created a hotter and drier environment around the fruit.
- Such conditions may increase fruit damage and rot risk.
- This may help explain the higher rotten rate under OTC.

Conclusions, Limitations, & Future Directions

Conclusions:

- Long-term heat experiment did not show significant results.
- Acute experiment significantly reduced yield and increased rot.

Limitations:

- Errors in sensors.
- Small number of biological replicates — limits power.

Future Directions:

- Increase N.
- Quality analysis based on gallic concentration.